

# Liquid SD Preparation

Liquid medium will be used for the majority of the project, since it allows for rapid growth and easy measurement via OD. This media lacks uracil supplementation needed for untransformed auxotrophic BY4741 *S. cerevisiae* to survive. Makes 1L of liquid synthetic defined / synthetic dextrose (SD) media.

## Reagents

| ✓                        | Name                                       | Quantity  | Location/Notes                                              |
|--------------------------|--------------------------------------------|-----------|-------------------------------------------------------------|
| <input type="checkbox"/> | Yeast Synthetic Drop-out Medium Supplement | 1.92g     | LACKS URACIL or LEUCINE<br>(e.g. Y1501 from sigma alderich) |
| <input type="checkbox"/> | Yeast Nitrogen Base without amino acids    | 6.7g      |                                                             |
| <input type="checkbox"/> | ddH2O                                      | 1L        |                                                             |
| <input type="checkbox"/> | D-glucose / dextrose                       | 20g       |                                                             |
| <input type="checkbox"/> | Sterile water                              | As needed |                                                             |

## Equipment

| ✓                        | Name             | Quantity | Location/Notes    |
|--------------------------|------------------|----------|-------------------|
| <input type="checkbox"/> | Analytical scale |          |                   |
| <input type="checkbox"/> | Autoclave        |          | Training required |

|                          |                            |  |                                |
|--------------------------|----------------------------|--|--------------------------------|
| <input type="checkbox"/> | Stir bar                   |  |                                |
| <input type="checkbox"/> | 500mL autoclaveable bottle |  | Used to make dextrose solution |
| <input type="checkbox"/> | 1L autoclaveable bottle    |  | Used to store media            |
|                          | BSC                        |  | Training required              |

## Instructions

1. Prepare the dextrose solution by adding the following to the **500mL bottle**, use a stir bar to stir:

| Reagent Tracker                               | Quantity |
|-----------------------------------------------|----------|
| <input type="checkbox"/> D-glucose / dextrose | 20g      |
| <input type="checkbox"/> ddH <sub>2</sub> O   | to 200mL |

2. Prepare the rest of the solution by adding the following to the **1L bottle**, use a stir bar to stir:

| Reagent Tracker                                                     | Quantity |
|---------------------------------------------------------------------|----------|
| <input type="checkbox"/> Yeast Synthetic Drop-out Medium Supplement | 1.92g    |
| <input type="checkbox"/> Yeast Nitrogen Base without amino acids    | 6.7g     |

|                                             |                |
|---------------------------------------------|----------------|
| <input type="checkbox"/> ddH <sub>2</sub> O | to 800 $\mu$ l |
|---------------------------------------------|----------------|

3. Autoclave both liquids at 121°C for 15min and allow them to cool
4. In a BSC, add the dextrose solution to the 1L bottle, add sterile water until the solution is 1L

#### Notes here

1. Can be stored in 4°C fridge for a few months [4] [5]
2. DO NOT RE-AUTOCLAVE (see below)
3. Solutions must be autoclaved separately and later combined to avoid triggering the maillard reaction in the autoclave , which could make toxic by-products and lower glucose concentration [3]

## References

[1] <https://bitesizebio.com/52937/yeast-media/>

[2]

<https://www.sigmaaldrich.com/CA/en/technical-documents/protocol/microbiological-testing/microbial-culture-media-preparation/yeast-drop-out#products>

[3]

[https://www.researchgate.net/post/Autoclaving\\_Glucose\\_in\\_yeast\\_peptone\\_Dextrose\\_plate\\_with\\_Agar](https://www.researchgate.net/post/Autoclaving_Glucose_in_yeast_peptone_Dextrose_plate_with_Agar)

[4]

<https://www.researchgate.net/post/How-long-can-I-store-un-poured-LB-YPD-agar-for-plates-in-duran-bottles>

[5] <https://shop.gmexpression.com/YPD-medium-liquid>